

Activity - 4

Input

Table Size = 10

Separate chaining

Key	h(k)
4371	1
1323	3
6173	3
4199	9
4344	4
9679	9
1989	9

collision

Occurs at 3.9

index	chain
0	-
1	4371
2	-
3	1323 → 6173
4	4344
5	-
6	-
7	-
8	-
9	4199 → 9679 → 1989

b) $h_1(m) = ((h(m)) + 1) \bmod 10$

$(6173 + 1) \% 10 = 4$ occupied

$(6174 + 1) \% 10 = 5$ free

$(9679 + 1) \% 10 = 0$ free

$(1989 + 1) \% 10 = 0$ occupied

$(1990 + 1) \% 10 = 1$ occupied

$(1991 + 1) \% 10 = 2$ free.

Final hash table

index	value
0	9679
1	4371
2	1989
3	1323
4	6173
5	4344
6	-
7	-
8	-
9	4199

(iii) Open addressing - Quadratic Probing

4371
 1 → index 1
 6173
 3 occupied
 $i = 1, 3 + 1^2 = 4$
 index 4.

1323
 2 → index 3
 4199
 9 → index 9

4344
 4 occupied
 $4 + 1^2 = 5 \rightarrow \text{index } 5$

9679
 9 occupied
 $9 + 1^2 = 0$
 → index 0

~~1989~~
 $i = 1$; occupied.
 $i = 2$; $(9 + 4) \bmod 10 = 3$ occupied
 $i = 3$; $(9 + 4) \bmod 10 = 3$ index 3.

index	value.
0	9679
1	4371
2	-
3	1323
4	6173
5	4344

and $h_2(n) = -1 - (n \bmod 1)$

The probing formula is

$$h_1(n) = h_1(n) + \phi \times h_2(n) \bmod 10$$

4371	1	5	2
1323	3	2	5
6173	3	6	1
4199	9	3	4
4344	4	4	3
9679	9	5	2
1989	9	1	6

Inserting

4 3 7 1

$h_1 = 1 \rightarrow \text{index } 1$

1 3 2 3

$h_1 = 3 \rightarrow \text{index } 3$

6 1 7 3

$h_1 = 3$ occupied

stop size $h_2 = 1$

$3 + 1 = 4$

index = 4

4 1 9 9

$h_1 = 9 \rightarrow \text{index } 9$

4 1 3 4 4

$h_1 = 4$ occupied

$h_1 = 3$

	Final	Percent
	index	value
2	0	-
1	4	4371
3	2	-
4	3	1323
5	4	6173
6	5	9679
7	6	-
8	7	4344
9	8	-
	9	4199

Arif